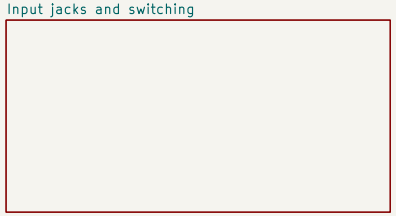
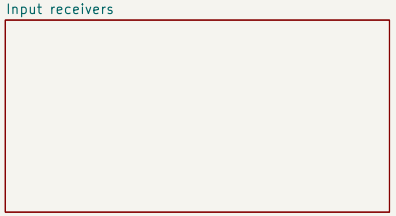
Plug & Roll main board. Each block of the board is one sub-sheet.

Input jacks and switching



File: input-jacks.kicad_sch

Input receivers



File: input.kicad_sch

INST buffer



File: input-inst.kicad_sch

Sheet: /
File: main-board.kicad_sch

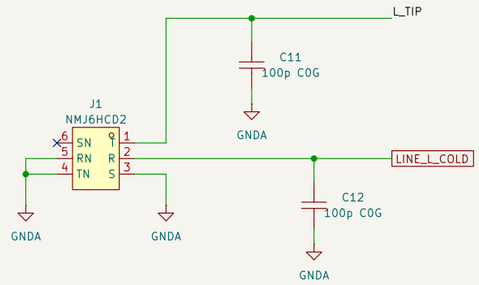
Title: Plug & Roll - Main board

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/4

INPUT JACKS

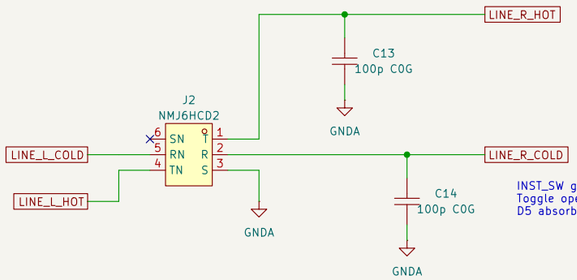
Tip = hot (+), ring = cold (−), sleeve = analog ground.
TN / RN / SN are switch contacts that rest on T / R / S while the jack is empty.

LEFT input



Empty jack: tip and ring are grounded (silent input).

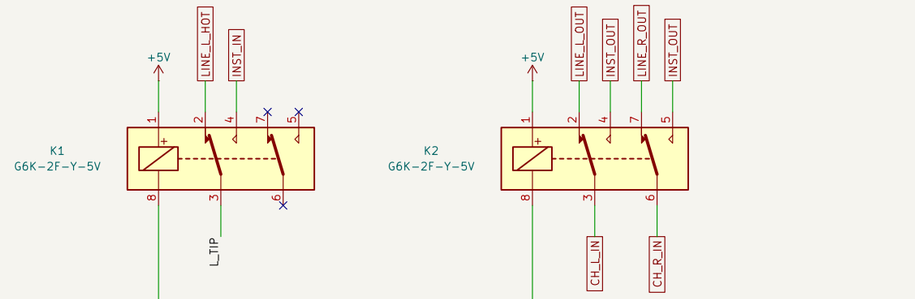
RIGHT input



Empty jack: the Right receiver gets the Left signal (mono).
Purely analog, the Daisy is not involved.

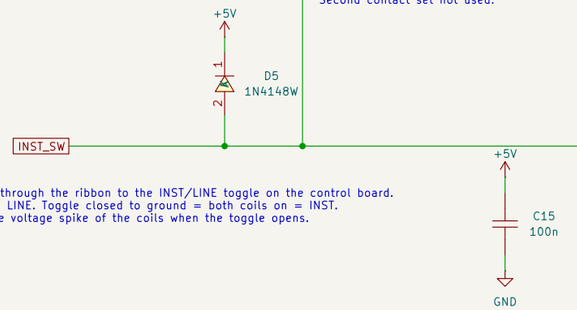
INST / LINE SWITCHING

Relays drawn released (no coil current) = LINE. Panel toggle closed = coils on = INST.



K1: Left tip to the LINE receiver or to the INST buffer.
Second contact set not used.

K2: each channel takes its LINE receiver or the INST buffer.

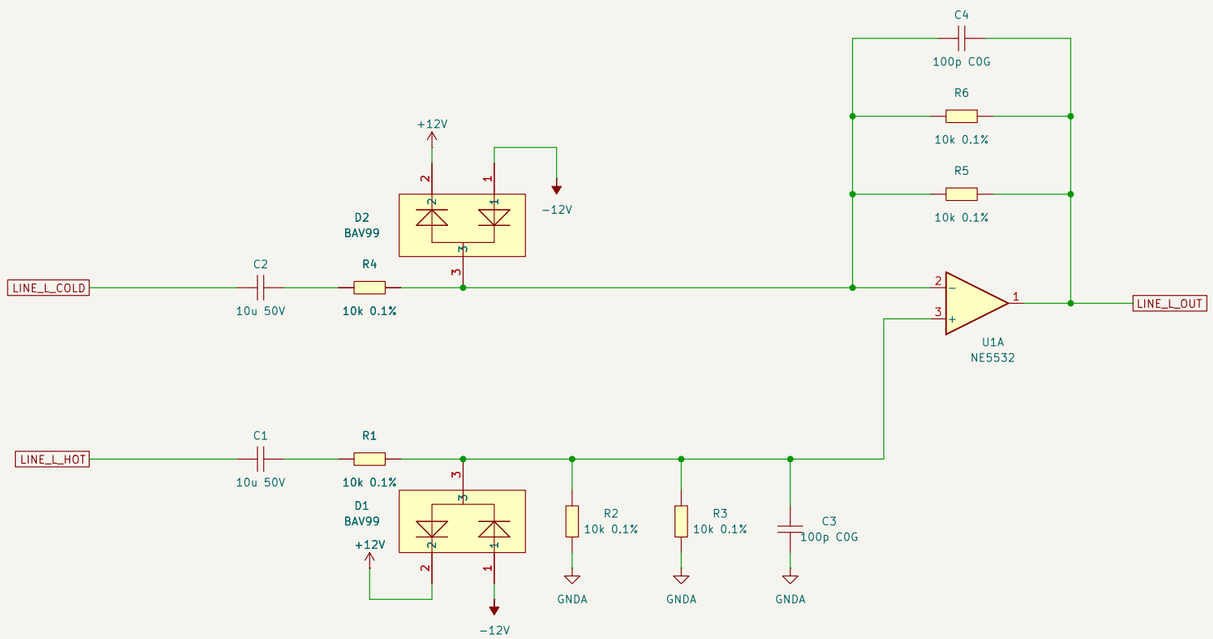


INST_SW goes through the ribbon to the INST/LINE toggle on the control board.
Toggle open = LINE. Toggle closed to ground = both coils on = INST.
D5 absorbs the voltage spike of the coils when the toggle opens.

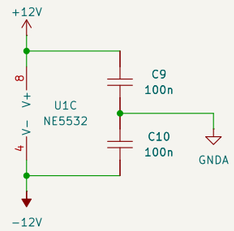
5 V decoupling, placed at the relays

Sheet: /Input jacks and switching/		
File: input−jacks.kicad_sch		
Title: Plug & Roll − Main board − Input jacks and INST/LINE switching		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/4

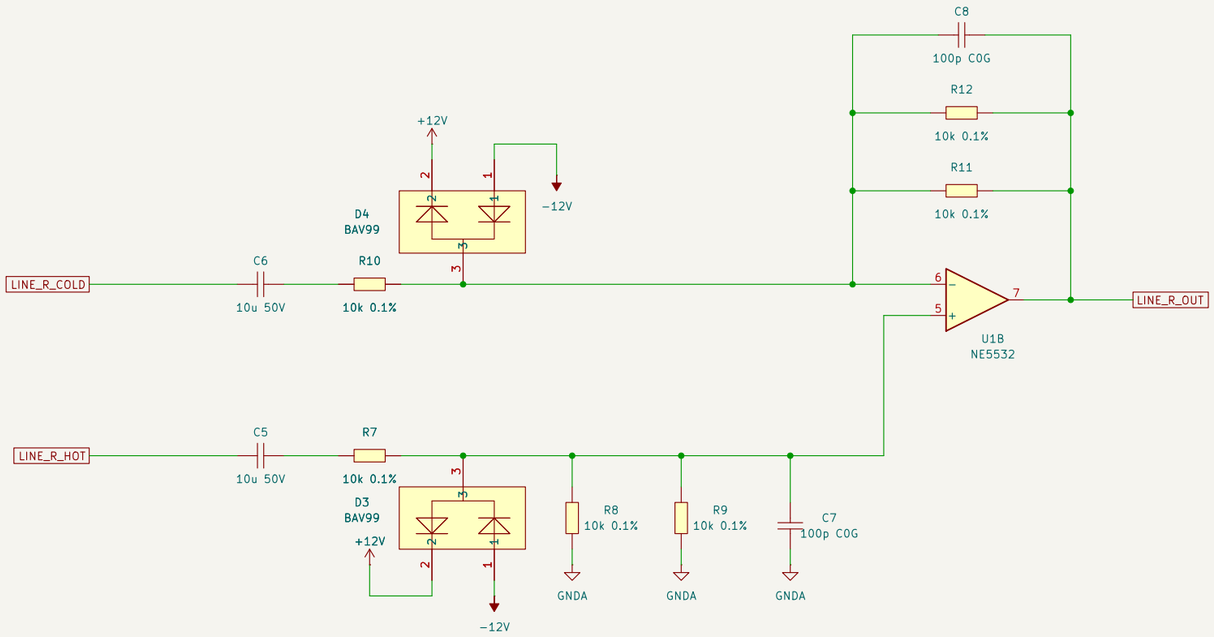
LINE receiver L: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



U1 supply and decoupling



LINE receiver R: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



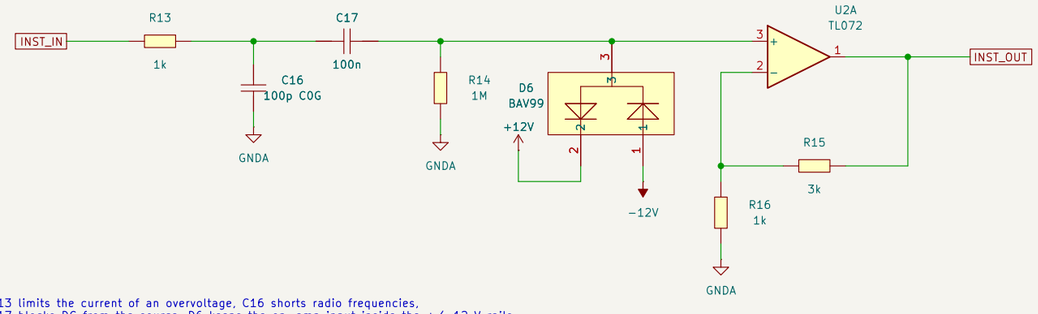
Sheet: /Input receivers/
File: input.kicad_sch

Title: Plug & Roll – Main board – Input receivers

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/4

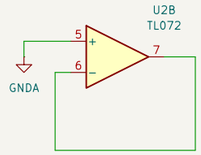
INST BUFFER

Instrument (guitar) input, used only in INST mode. Input resistance 1 Mohm, gain x4 (+12 dB) = 1 + R15/R16.

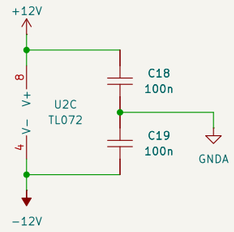


R13 limits the current of an overvoltage, C16 shorts radio frequencies,
C17 blocks DC from the source, D6 keeps the op-amp input inside the +/−12 V rails.

Unused half of U2



U2 supply and decoupling



Sheet: /INST buffer/
File: input-inst.kicad_sch

Title: Plug & Roll – Main board – INST buffer

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/4